WildEdge AI

Zero-Cloud Wildlife Monitoring System

Knowledge Distillation Pipeline — Proof of Work Report

Teacher Model: 95.0% accuracy (23344 KB)
Student Model: 79.8% accuracy (274 KB)
Compression: 85.1x smaller | Knowledge Retained: 84%

Animals: bear, deer, elephant, gorilla, leopard, lion, tiger, zebra
Dataset: 3360 training + 840 validation images
Target Hardware: ESP32-CAM (520KB RAM, 4MB Flash)

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Knowledge Distillation Pipeline Overview

PHASE 1: TEACHER MODEL

MobileNetV2 (α=1.0)
160×160 input
ImageNet pretrained
Fine-tuned on jungle animals

↓

Result: 95.0% accuracy
23344 KB (TOO BIG for ESP32)

Soft Labels
(T=5.0)

→

PHASE 2: STUDENT MODEL

Custom Depthwise-Separable CNN
96×96 input
5 conv blocks
GlobalAveragePooling

↓

Architecture: 10,832 params
Target: < 500KB after quantization

↓

PHASE 3: KNOWLEDGE DISTILLATION

```
Loss = α × KL(Teacher_soft, Student_soft) × T²
    + (1-α) × CrossEntropy(true_label, Student)
  Temperature: 5.0 | Alpha: 0.7 | Epochs: 80
```

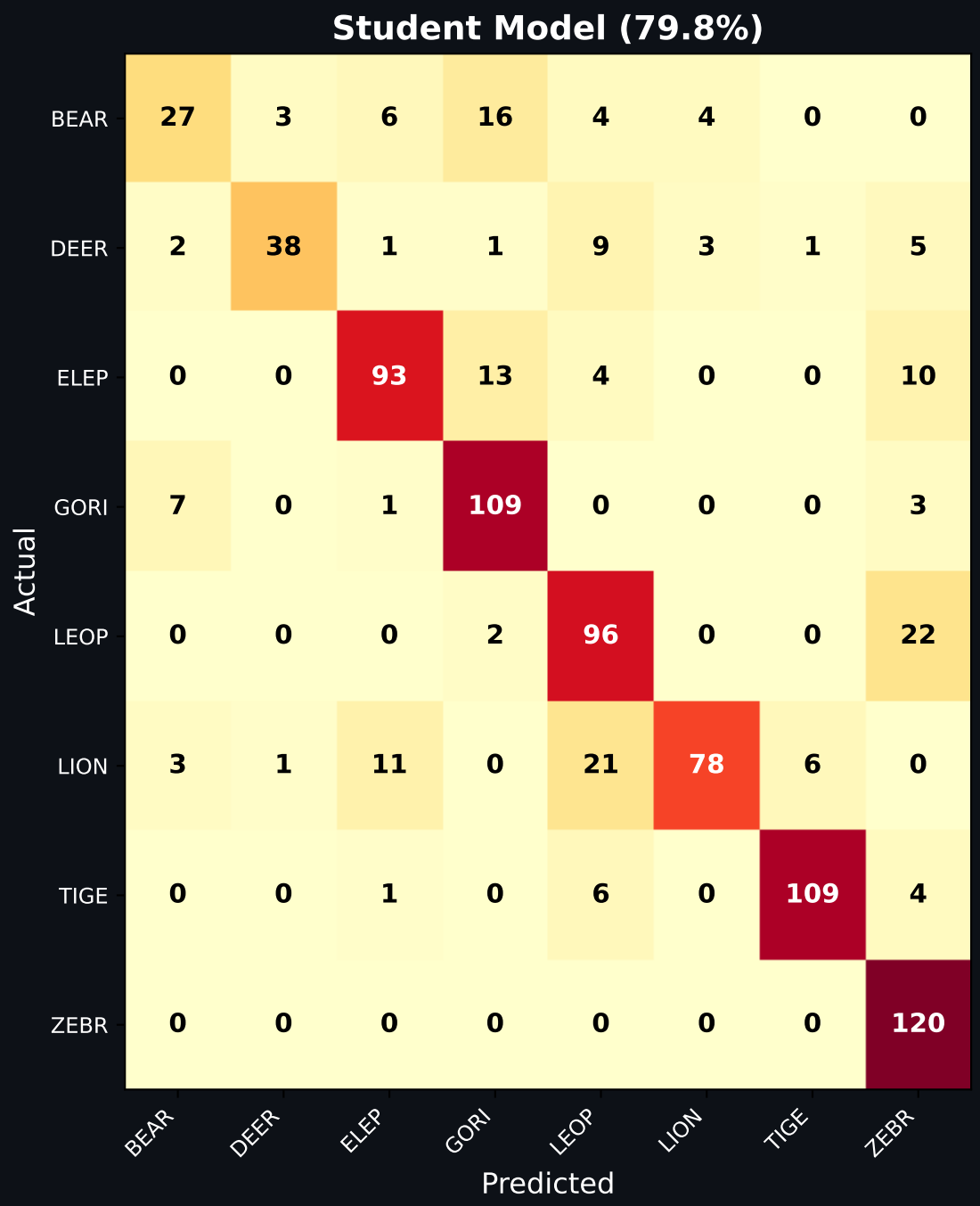
↓

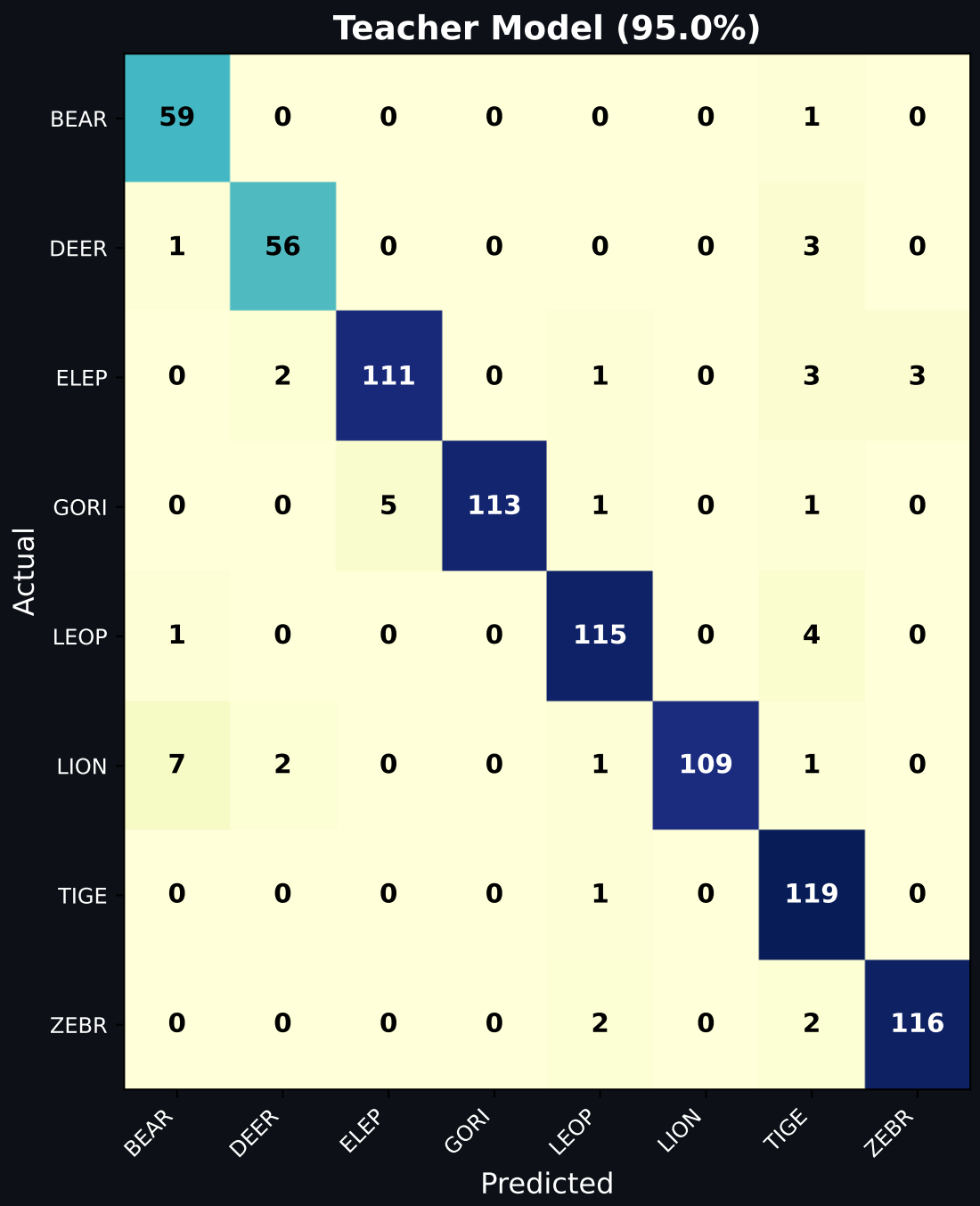
PHASE 4: FINE-TUNE + FINAL RESULT

```
 Student: 79.8% accuracy | 274 KB
 Compression: 85.1x smaller
 Knowledge retained: 84%
 Ready for ESP32-CAM deployment
```

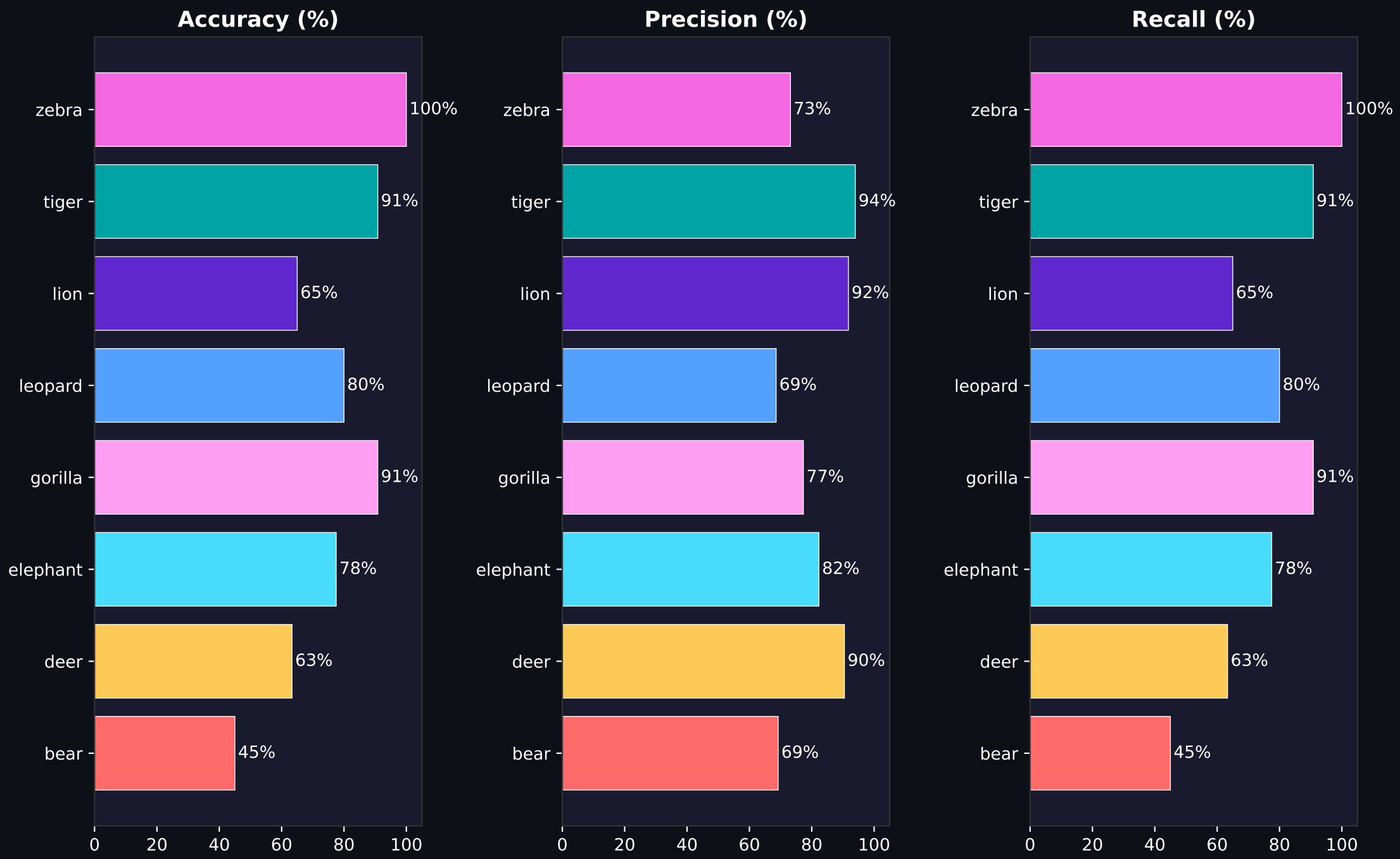
Dataset Overview

Confusion Matrices — Prediction Analysis





Per-Class Performance Metrics (Student Model)



Model Compression Analysis

Metric	Teacher Model	Student Model	Improvement
Architecture	MobileNetV2 (α=1.0)	Custom DSC-CNN	—
Input Resolution	160×160	96×96	—
Parameters	2,430,728	10,832	224.4x fewer
Model Size	23344 KB	274 KB	85.1x smaller
Accuracy	95.0%	79.8%	84% retained
After INT8 Quantization	N/A (not needed)	~69 KB (est.)	**Fits ESP32**
Inference Device	PC/Server	ESP32-CAM ($6)	—
Cloud Required	Yes	No (fully offline)	
Power Consumption	~100W (GPU)	~0.5W (MCU)	200x less

Student Model — Sample Predictions on Validation Set

✓ zebra
(100%) [act: zebra]

✓ leopard
(48%) [act: leopard]

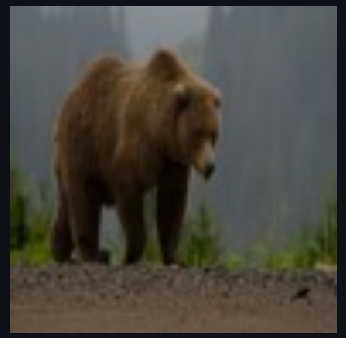
✗ elephant
(48%) [act: lion]

✓ lion
(89%) [act: lion]

✓ gorilla
(97%) [act: gorilla]

✓ lion
(47%) [act: lion]

✗ deer
(83%) [act: bear]



✗ lion
(83%) [act: deer]

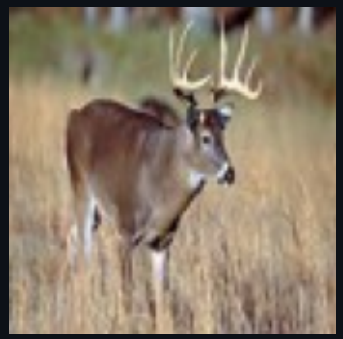
✓ tiger
(97%) [act: tiger]

✓ gorilla
(96%) [act: gorilla]

✓ leopard
(100%) [act: leopard]

✓ elephant
(96%) [act: elephant]

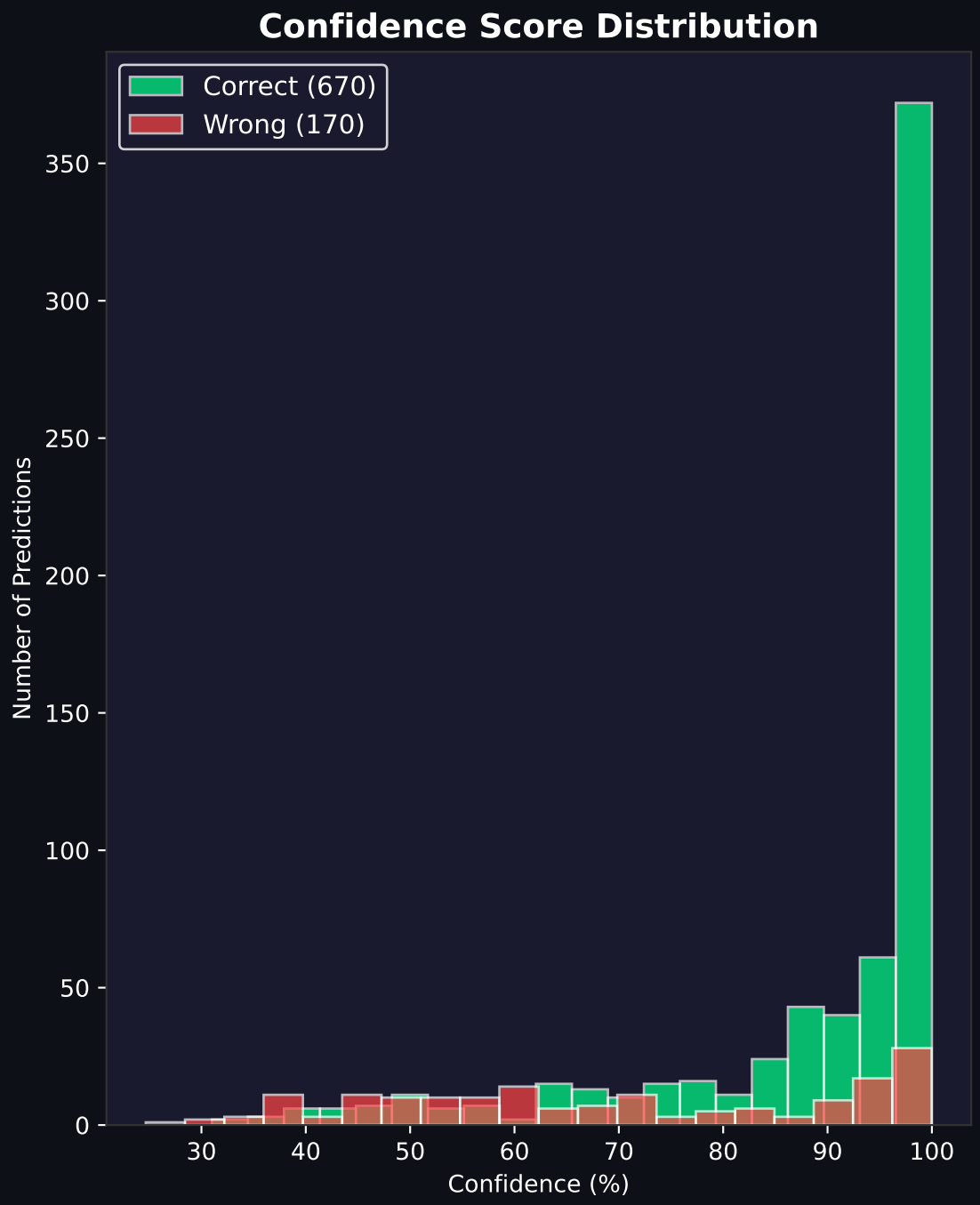
✓ deer
(75%) [act: deer]

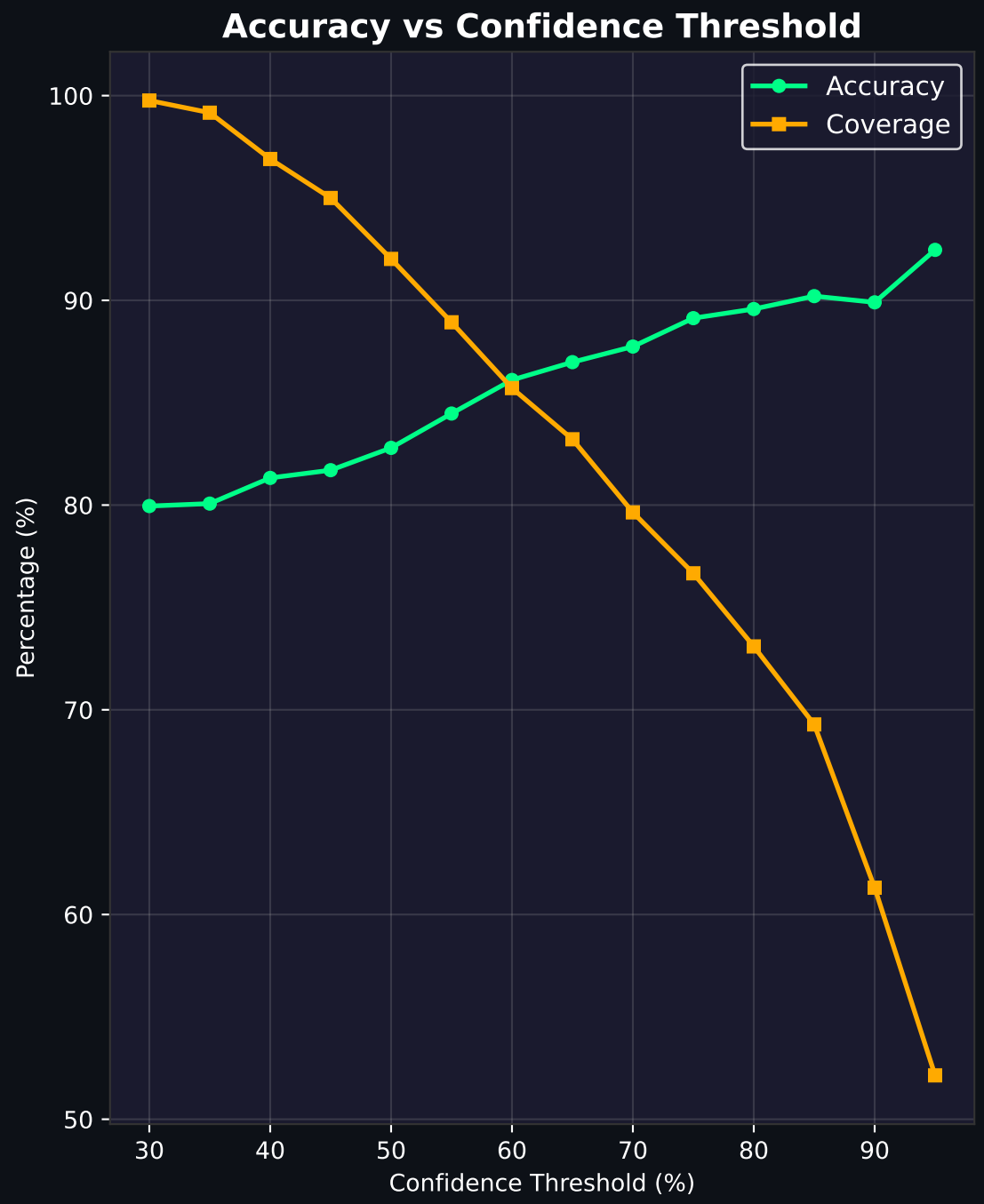


✓ tiger
(47%) [act: tiger]

✓ lion
(87%) [act: lion]

Prediction Confidence Analysis





ESP32-CAM Deployment Specifications

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│                          TARGET HARDWARE: ESP32-CAM                      │
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│                                                                          │
│   Microcontroller:    ESP32-S (Xtensa LX6, Dual-Core)                    │
│   Clock Speed:        240 MHz                                            │
│   SRAM:               520 KB (usable ~200 KB for ML)                     │
│   PSRAM:              4 MB (external, for camera buffer)                 │
│   Flash:              4 MB (stores code + model)                         │
│   Camera:             OV2640 (2MP, RGB565 output)                        │
│   WiFi:               802.11 b/g/n (for alert transmission)              │
│   Power:              3.3V / ~200mA active                               │
│   Cost:               ~$6 USD / ₹500 INR                                 │
│                                                                          │
├──────────────────────────────────────────────────────────────────────────┤
│                          DEPLOYED MODEL SPECS                            │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│   Model:              Knowledge-Distilled DSC-CNN                        │
│   Parameters:         10,832                                             │
│   Size (Keras):       274 KB                                             │
│   Size (INT8 TFLite): ~69 KB (estimated)                                 │
│   Input:              96×96×3 RGB image                                  │
│   Output:             8-class softmax                                    │
│   Classes:            bear, deer, elephant, gorilla, leopard, lion, tiger, zebra  │
│   Accuracy:           79.8% (validated)                                  │
│   Est. Inference:     ~500ms - 2000ms per frame                          │
│   Framework:          TensorFlow Lite Micro (C++)                        │
│                                                                          │
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│                          SYSTEM FEATURES                                 │
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│                                                                          │
│   □ Fully offline — zero cloud dependency                                │
│   □ Real-time inference on microcontroller                               │
│   □ Solar-powered capable (~0.5W consumption)                            │
│   □ WiFi/LoRa alert when dangerous animal detected                       │
│   □ <₹1000 per node — mass deployable                                    │
│   □ Weather-proof enclosure compatible                                   │
```

Student Model — Layer Architecture

batch_normalization	(None, 96, 96, 16)	64
re_lu	(None, 96, 96, 16)	0
max_pooling2d	(None, 48, 48, 16)	0
depthwise_conv2d	(None, 48, 48, 16)	144
batch_normalization_1	(None, 48, 48, 16)	64
re_lu_1	(None, 48, 48, 16)	0
conv2d_1	(None, 48, 48, 24)	384
batch_normalization_2	(None, 48, 48, 24)	96
re_lu_2	(None, 48, 48, 24)	0
max_pooling2d_1	(None, 24, 24, 24)	0
depthwise_conv2d_1	(None, 24, 24, 24)	216
batch_normalization_3	(None, 24, 24, 24)	96
re_lu_3	(None, 24, 24, 24)	0
conv2d_2	(None, 24, 24, 32)	768
batch_normalization_4	(None, 24, 24, 32)	128
re_lu_4	(None, 24, 24, 32)	0
max_pooling2d_2	(None, 12, 12, 32)	0
depthwise_conv2d_2	(None, 12, 12, 32)	288
batch_normalization_5	(None, 12, 12, 32)	128
re_lu_5	(None, 12, 12, 32)	0
conv2d_3	(None, 12, 12, 48)	1,536
batch_normalization_6	(None, 12, 12, 48)	192
re_lu_6	(None, 12, 12, 48)	0
max_pooling2d_3	(None, 6, 6, 48)	0
depthwise_conv2d_3	(None, 6, 6, 48)	432
batch_normalization_7	(None, 6, 6, 48)	192
re_lu_7	(None, 6, 6, 48)	0
conv2d_4	(None, 6, 6, 64)	3,072
batch_normalization_8	(None, 6, 6, 64)	256
re_lu_8	(None, 6, 6, 64)	0
global_average_pooling2d_1	(None, 64)	0
dropout_2	(None, 64)	0